

Element D:

D1: Our process for generating and comparing possible designs for our aquaponics system went through multiple steps, including researching different designs that were already proven to work, altering those designs to fit our criteria, and then concluding if those designs are worthy of pursuing or not. By considering factors such as space constraints and resource availability, we were able to determine the most suitable design for further development. We then decided upon the pros and cons of each design, as well as which tradeoffs are beneficial and which are not necessary. By analyzing each of the designs, we then combined multiple together, creating one design encapsulating all the pros and less of the cons of the combined ideas.

With this process in mind, we made 4 concepts that could possibly be a viable solution that satisfies the design requirements most efficiently. See the table below for a detailed listing of

Conceptual designs

Design Requirement	Concept 1 (Linear)	Concept 2 (Modular)	Concept 3 (Casade)	Concept 4 (Minimalistic)
Educational Value	Easy to understand and go through with the kids	Compacted enough to be placed inside classrooms for more controlled visited and education	Demonstrates new and creative ideas for a common problem	Bare bones and easy for kids to copy and learn from.
Scale and Size	Large length Short width Low height Fits inside the area	Short length Short Width Tall height Fits inside the area	Large length Large length Tall Height Fits inside the area	Short length Short Width Tall height Fits inside the area
Safety Considerations	Low risk of collapsing and leakage would be minimal	Less surface area for kids to trip on	Easy for the kids to feed the fish without problems	Less surface area for kids to trip on
Accessibility	Easy to reach plants and fish-tank.	Easy to reach plants and fish-tank	Easy to reach plants and fish-tank	Easy to reach plants and fish-tank
Sustainability	Can have the ability to sustain	Can have the ability to sustain	Can have the ability to sustain	Can have the ability to sustain

	fish and plants	fish and plants	fish and plants	fish and plants
Monitoring and Control	Flat layout allows for easier maintenance	Simple fish tank below and easy to access plant shelf allows maintenance easier	The front fish tank exposed allows for better monitoring of the fish tank. Pumps and tubes are easy to see on the back.	Simple fish tank below and easy to access plant shelf allows maintenance easier

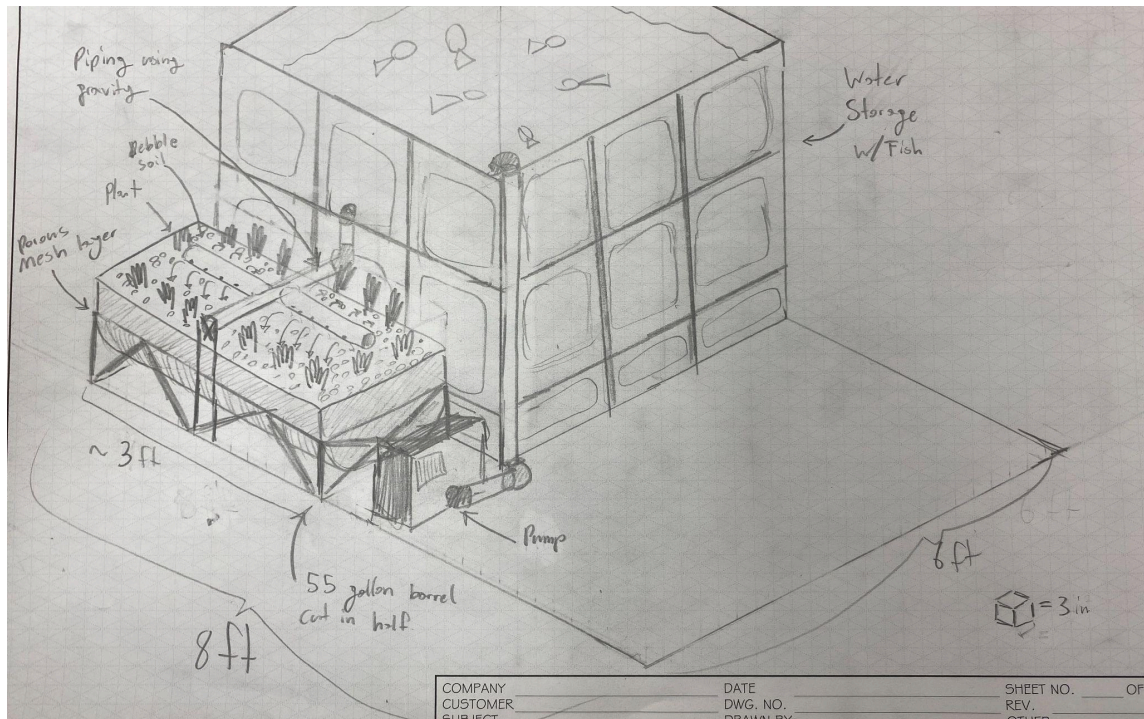
D2:

	Easy to Build	Creative	Probability of Success	Cost	Food Production	
Modular Design	2	3	4	3	3	3.2
Linear Design	3	2	4	2	2	2.85
Cascade Design	2	4	3	2	3	2.65
Minimalistic Design	4	2	4	3	4	3.65
Percent -	15%	5%	35%	25%	20%	

We decided to combine our two highest scoring designs, the modular design and the minimalistic design. By combining these two, we can take the best parts of each of the designs and make them into one. We will have the fish tank on a higher elevation than the grow beds, and the water will be moved with gravity down through the grow beds, which gets filtered through the clay pebbles so the nutrients stay in the grow beds with the plants, until the filtered water gets pumped back up into the fish tank. The top of the fish tank opens in order to provide the fish with food.

D3: After researching and designing our chosen conceptual design, these are the next steps necessary to build a successful aquaponics system. First, we need to acquire the needed materials and organisms for the system. This will happen by using the two thousand dollars given to our group from Millersville University. Next, construction will begin in the middle school. By following our design, the construction period should be easy because we are simply turning our design from a conceptual design to reality. After constructing the system, we will implement the plants and fish. After the implementation of the organisms, we will measure the production of the plants

D4:



Item	Price Per Unit in US \$	Quantity Needed	Overall Cost in US \$	Link to website or store where Purchased
Fish (Tilapia)	\$2.50	15	\$37.50	https://liveaquaponics.com/collections/tilapia/products/mozambique-tilapia?variant=32672166674567
Plant Seed	\$9.98	1	\$9.98	Sereniseed Certified Organic Leafy Greens Lettuce Seeds Collection (10-Pack) – 100% Non GMO, Open Pollinated – Grow Guide
Plumbing: AquaParts SS2 Kit	\$249.95	1	\$249.95	https://www.theaquaponicsource.com/shop/plumbing-parts-kits/aquaparts-plumbing-kits/aquaparts-ss2-aquaponics-plumbing-kit/
Getting Started Kit	\$139.95	1	\$139.95	https://www.theaquaponicsource.com/shop/cycling-kits-and-bacteria/cycling-kits/aquastart-aquaponics-getting-started-kit/
Clay pebbles for the growing bed	\$42.99	2	\$85.98	https://www.theaquaponicsource.com/shop/grow-beds-media/growing-media/growt-clay-pebbles-grow-media-4-mm-16-mm-40-l/
Plant Growth Lights	\$79.99	1	\$79.99	https://www.walmart.com/ip/seort/976449692
Pump	\$75.95	1	\$75.95	https://www.theaquaponicsource.com/shop/pumps/submersible-pumps/active-aqua-submersible-pump/

One and a Half Inch Diameter PVC Piping	\$12.78	2	\$25.56	https://www.homedepot.com/p/Charlotte-Pipe-1-1-2-in-x-10-ft-PVC-Schedule-40-DWV-Pipe-PVC071120600/100348474
1-1/2 in. PVC Sch. 40 Solvent x Solvent Ball Valve	\$11.82	1	\$11.82	https://www.homedepot.com/p/Everbilt-1-1-2-in-PVC-Sch-40-Solvent-x-Solvent-Ball-Valve-107-637EB/319340971
Total	\$625.91	25	\$716.68	